



Reflective Design Theorizing with User Interviews: A Case Study for AI Energy Labels

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Abstract. Several information technology (IT) artifacts exist that can benefit sustainable development, however were not developed under adequate consideration of design science research methodology. In response, we present a methodological contribution by introducing a qualitative reflective theorizing approach that derives design principles from already existing IT artifacts with grounded theory procedures. We validate our approach by investigating the special case of communicating properties of artificial intelligence (AI) models via high-level consumer labels, drawing inspiration from well-established systems such as energy labels and Nutri-Score. Such AI energy labels inform diverse stakeholders on the predictive quality and resource consumption of models in comprehensible ways, however design knowledge has not been properly captured for their development. Building on existing reflective practices like design archaeology, we conceptualize how adequate design principles can be developed based on existing artifacts in-situ, grounded in results from a qualitative interview study. Our research offers insights into reflective design processes and proposes four actionable design principles for advancing transparency and sustainability in AI.

Keywords: Reflective design theorizing · design principles · AI energy labeling · sustainability

1 Introduction

The growing emphasis on sustainability has spurred the development of numerous IT artifacts aimed at supporting environmentally conscious practices, such as software eco-labels [7] and artificial intelligence (AI) energy labels [8]. Many of these artifacts were constructed without explicitly adhering to design science

research (DSR) processes. While they are undoubtedly valuable in promoting sustainable development [9], such artifacts could be made even more effective and disseminable by explicating the design knowledge that they inherit.

Recognizing the opportunity to extract design knowledge from DSR-agnostic artifacts, purely *reflective* DSR studies have emerged as a growing trend. These studies focus on synthesizing design theory from completed artifacts and independently of the artifact building project, allowing design researchers to *theorize* from DSR-agnostic artifacts [31]. Previous methodological work on reflective DSR studies prescribes the analysis of completed IT artifacts *in-situ*, meaning in an authentic use context with a realistic user base [5]. However, in many existing reflective studies, design researchers frequently analyze IT artifacts by becoming users themselves, systematically interacting with the artifact and reflecting on its components [23, 35]. The former approach risks analyzing the artifact outside of its realistic use context, or *ex-situ*, as design researchers might not be the intended user group of the artifact [22].

To illustrate the limitations of ex-situ reflective studies, we turn to the design of energy labels. In various forms, these labels are mandated globally with the intent to provide consumers of electronic appliances with essential information about long-term costs and environmental impact. From a DSR perspective, these labels may inherit meaningful design knowledge that can be explicated and generalized to inform the design of labeling practices in other fields. While researchers can contemplate on label design based on their own experiences, this approach is potentially subjective and limited. Further, observing customers glancing at labels in a store may reveal behavioral patterns, but it fails to provide deeper insights on how the design of the labels is perceived. In response, qualitative methods like interviews have a high potential for dismantling IT artifact’s underlying design properties and generating actionable design theory. To underpin this hypothesis, we seek to answer the following research question:

RQ 1. *How can user interviews on existing IT artifacts inform reflective theorizing in DSR?*

Naturally, validating any possible answer to RQ1 requires a practical investigation, where reflective user-involved theorizing is applied to a meaningful, DSR-agnostic IT artifact. Due to their promising benefits for improving the sustainability of AI [9], we study *energy labels for AI models* [8] as suitable objects for design theorizing. The labels in question were designed in analogy to the EU energy labels [8] and without explicitly considering DSR methodology and, to the best of our knowledge, no further existing theory of action and design was derived. However, a recent interview study evaluated AI energy labeling practices [10], and based on these results we therefore seek to answer:

RQ 2. *What are design principles for AI energy labels?*

For answering RQ1, we propose to analyze qualitative interview results using grounded theory methodology, which facilitates inductive generation of theory from qualitative data [33]. The answer to RQ2 is the logical product of applying this reflective theorizing method to the AI energy labeling interview analysis

data. In our theorizing process, we synthesize prescriptive knowledge comprising four design principles (DPs) that reinforce the use of AI energy labels for practitioners. The research process at a glance is depicted in Fig. 1.

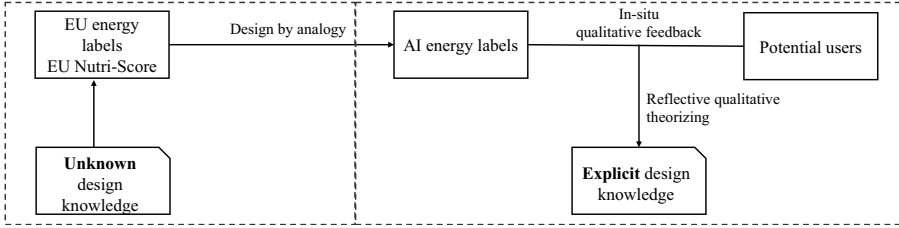


Fig. 1. The research process for qualitative reflective theorizing on AI energy labels.

The remainder of the paper is organized in four sections. First, we establish the research context for this work by discussing literature on theorizing in DSR, reflective theorizing approaches, grounded theory as well as energy labeling in the context of AI sustainability. Afterwards, the study’s research design is presented in detail. Third, we present the DPs derived for AI energy labels. We conclude our work with a critical discussion of our findings, formulating implications for design research alongside the study’s limitations and potential for future work.

2 Research Context

2.1 Design Theorizing and Logical Reasoning

The primary goal of design science research (DSR) is to accumulate design knowledge, which serves as a mechanism to generalize insights from specific design instances to broader application scenarios [21]. Design knowledge, in essence, refers to “knowledge that can be used to produce designs” [1]. This knowledge can be represented in various ways, most prominently by *artifacts* or with *prescriptive knowledge*, commonly categorized under design and action *theory* [11, 12]. Prescriptive knowledge on how to do something to achieve specified goals, is frequently codified as DPs.

There is limited methodological work in which the process of prescriptive knowledge generation is observed detached from the overall DSR process. For the sake of consistency, we will refer to the generation of prescriptive knowledge as *theorizing*. To characterize the DP theorizing process, Purao et al. [22] provide four dimensions of DP generation. (1) DPs are *influenced* by prior theories, design processes, and completed artifacts and evaluations. (2) *Temporality* refers to when DPs are developed, whether during design-evaluate cycles, post-deployment, or retrospectively from past artifacts. (3) *Actors*, such as the original DSR team, new researchers, or domain experts, each bring unique advantages, but also risks like over-familiarity or lack of context. (4) *Content* involves the

form of DP documentation, aiming to balance practical usability for practitioners with precise knowledge accumulation for the DSR community.

We characterize our research along these four dimensions. Regarding dimension 1, our proposed theorizing method generates DPs influenced by a completed artifact as well as interviews with artifact users, which can be viewed as evaluative interviews. In line with dimension 2, we develop our DPs retrospectively, in accordance with other reflective theorizing studies. Regarding dimension 3, we bring in the perspective of domain experts alongside our perspective as new design researchers. In the 4th dimension, we make use of existing conceptualizations about DPs. These provide guidance on DP documentation and formulation [4, 12]. Particularly, Chandra et al. [4] proposes three key components of a DP: action, referring to what an artifact enables users to do; materiality, describing how the artifact should be constructed to achieve the design goal; and boundary conditions, specifying the relevant use context or intended user group for which the design works effectively.

Other methodological work on theorizing focuses on its underlying *reasoning*. Theorizing procedures can be deductive, abductive, or inductive. Deductive theorizing is based on existing theory or tacit knowledge. Which is applied to a specific design problem to synthesize design knowledge [13]. In abductive approaches, existing theory is used creatively to find the most plausible, yet falsifiable explanation for an observation, colloquially referred to as *guessing* [6]. Generally, abduction allows researchers to derive principles, such as the notion that an artifact design should include a specific property, based on observations that a problem was consistently solved when that specific property was incorporated.

Induction is the third type of reasoning that is frequently used in theorizing. Inductive generation of design knowledge is purely data and observation driven. Observations in the DSR realm can be made via instantiation of design knowledge into IT artifacts and subsequent application in socio-technical systems [13, 16]. For instance, in action design research, DPs are frequently instantiated and evaluated via field experiments and interventions in real world organizations [1, 27]. Observed DPs and actions that are proven effective in practice can be explicated into design knowledge, following inductive reasoning approach. As we motivated in our introduction, we aim to study IT artifacts in-situ with the help of qualitative data from user interviews, we therefore propose an inductive way of theorizing that builds prescriptive knowledge without the need for existing theory or prior tacit knowledge.

2.2 Reflective Theorizing and Design Archaeology

Early DSR studies primarily contributed to design knowledge via artifact building [31]. With the establishment and popularization of design theory [11, 12], DSR projects increasingly advanced the cumulative body of knowledge with theoretical contributions like DPs that are tied to artifacts and their building process. Most recently, purely *reflective* DSR studies emerged as a new type of DSR study. These studies focus on synthesizing design theory from completed

artifacts and independently of artifact building projects [31], opening up more research opportunities for design researchers.

Reflection in design theorizing is twofold. On one hand, it denotes the design researchers' introspective contemplation on remembered design activities. On the other hand, reflection can be conducted on the IT artifact itself. As we motivated in our introduction, we will focus on reflection to synthesize knowledge from design artifacts rather than design activities. This allows for theorizing of DSR-agnostic artifacts, that is useful artifacts that were built without following a rigorous DSR methodology [5].

A unique angle on reflective design theorizing is taken on by *design archaeology* proposed in Chandra Kruse et al. [5], meaning the study of design activity by recovering and analyzing digital artifacts. Notably, design archaeology is conducted with artifacts ex-post facto, or after the design artifact has been built to completion. A key difficulty when performing design archaeology is analyzing the artifact in-situ, in its context of usage while analyzing the artifact through the lens of a designer. With the toolkit of reflective causal analysis, design archaeologists attempt to gain design knowledge without familiarity of the design process behind the artifacts. In this process, artifacts' impacts that go beyond their initial intended purpose are uncovered.

Specifying the unit of analysis for reflective theorizing is relevant to the reflective theorizing process [5]. It is evident that IT artifacts should best be analyzed in-situ to attend to the context specificity of digital artifacts. However, *existing design archaeology studies* use manifold units of analysis that partially lack involvement of users outside of the narrow team of design researchers. Recker [23] applies deductive, reflective theorizing to generate DPs for publicly available corona dashboards by analyzing the dashboard elements under the conceptual lens of representation theory, mapping each element onto a representation of a focal real-world phenomenon. In a similar way, Wulfert et al. [35] explores relevant e-commerce platform characteristics.

There are existing design archaeology studies which successfully involve users for theorizing. For instance, Stählin et al. [28] analyzes stakeholder interviews to generate design knowledge on how to empower health workers with mobile health. MacNeil et al. [19] surveys teams of design students after using online white boards for their work to derive in-situ design knowledge about online collaboration in white boards. This work goes into the same direction as our approach, however both papers do not generate DPs and give no specification about methods used for qualitative data analysis.

2.3 Grounded Theory Methodology for Design Theorizing

Grounded theory is an evolving methodology that is frequently used for constructing theories through qualitative data analysis. Originating in social science research, there has been a growth in popularity within the IS research field. Grounded theory allows researchers to “develop a theoretical account of the general features of a topic, while simultaneously grounding the account in empirical observations of data”. The methodology is effective at contributing

theory-constructs, that is the definition of variables, their relationships as well as justifications and boundary conditions for these relationships. It offers a collection of procedures that can be used for data collection and analysis [14, 33]. In the following, we present the three types of coding that we used for the analysis of the AI energy labeling interview study.

Open Coding: The initial stage of qualitative data analysis, where the researcher assigns labels to segments of data to identify distinct *concepts*. This process involves constant comparison of textual indicators, grouping similar words and phrases into broad thematic domains.

Axial Coding: With this type of coding, the researcher organizes and refines codes by identifying relationships between them and grouping them into categories or *themes*. This process involves constant comparison and the integration of inductive or deductive reasoning. Axial codes form the basis of theoretical development.

Selective Coding: Selective coding is the final stage of qualitative data analysis, where the researcher selects a core category and integrates it with other categories from axial coding to form a coherent and comprehensive narrative. This process unifies the key themes to construct meaning in the form of *dimensions*, allowing for the development of a theory or case story that illustrates how the interconnected concepts relate to the research findings.

Grounded theory is rarely used in existing DSR studies due to dissimilarities that have so far made the two methodologies seem incompatible. For one, DSR is motivated by solving meaningful problems, *changing* the world, while grounded theory methodology tends to provide theory that *fits* with the world that is grounded. Secondly, grounded theory aims to be more substantive, while traditional artifact-focused DSR projects often seek local solutions that are not always generalizable [14].

In spite of these dissimilarities and because we explore a methodically under-explored area of DSR, that is qualitative reflective design research, we seek to put grounded theory to the test due to key synergies that we identified. As previously mentioned, this study’s goal is to contribute to DSR methods with qualitative reflective theorizing. We view grounded theory as a suitable methodology for developing robust prescriptive knowledge from qualitative data due to its inductive nature, requiring no external theory or tacit knowledge about the analyzed design artifact.

2.4 AI Sustainability and Energy Labels

AI contains a dichotomy when it comes to sustainability and environmental impact. On the one hand, AI can be used in positive ways to advance sustainable development [15] and enable the efficient use of resources, for example by optimizing supply chains or reducing energy waste. On the other hand, AI advances follow a “bigger-is-better paradigm” [30] and as a result significantly impact our environment. This problem is likely to scale further with the growing

accessibility of AI [26], which places the sustainability of AI at the core of current research on AI ethics [29]. To illustrate this with some numbers, consider today’s popular generative AI models, with ChatGPT alone being reported to handle billions of user queries per day [25]. A recent study found that using such models quickly consumes the equivalent energy of charging smartphones, with each single query emitting up to one gram of CO₂ [17]. As a result, experts agree that the environmental costs of modern AI are a prime systematic risk to our world and demand more resource awareness [3].

For making AI development more sustainable, documentation and transparency were argued to be key criteria [24], requiring to inform “stakeholders from various facets of the AI lifecycle” [2]. Unfortunately, the current reporting overly focuses on predictive model capabilities and is mostly directed at developers and experts [9]. Well-established labeling systems, such as energy labels for devices or nutritional labels for food products, are however effective in raising user awareness about resource efficiency [32]. This fueled the push toward establishing an analog system in the context of using and developing AI models [8]. The proposed methods for sustainable and trustworthy reporting manifested this idea further [9]. The corresponding STREP software framework provides users not only with the opportunity to investigate the efficiency trade-offs of various models, but also allows them generate high-level labels as shown in Fig. 2. The works that developed AI energy labeling merely took inspiration from existing systems and were consistent in a DSR-agnostic design process, hence representing a good use case for testing reflective qualitative theorizing.

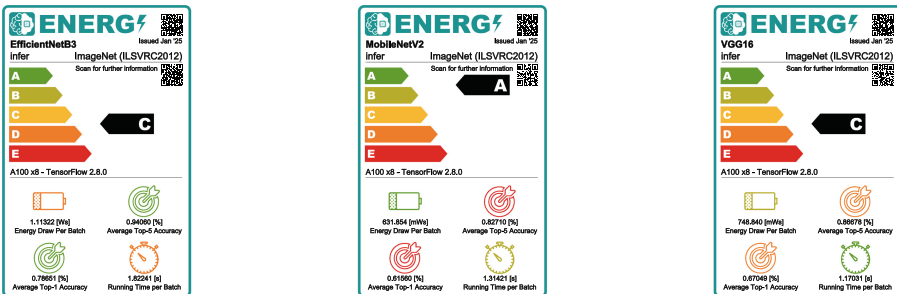


Fig. 2. AI energy label examples, generated with the STREP software [9]. The labels contain information on the tested AI model as well as training data and hardware environment used for testing. They also feature four singular performance and energy consumption metrics as well as a central, aggregative compound score.

3 Research Design

In order to answer RQ1, we explore whether grounded theory can be leveraged to derive DPs based on qualitative data about existing IT artifacts. For that, we

look at the specific case of AI energy labels, for which no DPs were yet derived, however qualitative feedback was collected in a recent interview study [10], which we analyze to answer RQ2.

3.1 Phase 1: Interview Study

Our study is based on data gathered from semi-structured interviews with participants recruited through a public call on social media and mailing lists. In total, 16 practitioners were interviewed, with varying levels of self-assessed AI expertise, namely beginner level (1), users (4), novel developers (8), and experts (3). Their educational backgrounds are equally diverse, with highest graduation levels spanning from high school diplomas to doctoral degrees. Interviews were conducted via Zoom with prior written consent for anonymized audio recordings. The study materials and results were also made publicly available to foster transparency and reproducibility [10].

In the interviews, participants were initially asked to introduce themselves and talk about their daily work routines involving AI, as well as challenges faced when developing or using AI models. After setting and recalling the work context, participants were presented with two prototype AI energy labels for image classification models generated with STREP [9]. The interviewees were asked to describe and compare their features, identifying strengths, areas of confusion, and potential improvements. Subsequently, they were presented with other formats that communicate AI model information, such as research papers, model cards [20], and fact sheets [18], discussing the practical relevance for their work. The final interview part discussed trust and credibility in AI energy labeling [34], focusing on certification, regulation, and potential authorities responsible for the labeling process. The interviewers formulated broad and open-ended questions and did not explicitly ask for any design knowledge or feedback, which makes this study a suitable data base for creative, grounded theorizing.

3.2 Phase 2: Reflective Theorizing Using Grounded Theory

After acquiring the interview transcripts of the original study, we analyzed them using MAXQDA 24.7, making use of the **Creative Coding** functionality that facilitates grounded theory procedures. Our goal was to derive design knowledge in the form of DPs from the interviewees' impressions of the AI energy labels. Initially, we applied open coding by identifying meaningful sections of the transcriptions and marking them with descriptive labels. As indicators, we focused on different types of reactions with respect to AI energy labels' various aspects and components. For example, we identified *confusion*, *misunderstanding*, *skepticism*, *positive/negative feedback*, and *wishes for improvement*. Additionally, we coded for potential AI energy label use cases that the interviewees conceived.

In the second iteration of open coding, we identified similarities as well as differences between the codes, synthesizing first order concepts which aim to summarize different indicators based on the aspect of the label that they are directed toward. This round of coding resulted in 21 *concepts*.

Subsequently, we applied axial coding with the intent to distill different fundamental *themes* that underlie a group of concepts. Themes followed one of the following three patterns (1) a limitation, downside, or source of confusion, (2) a wish, need, or suggestion on how to improve the label, or (3) a common theme describing the use context of the label. We used a partially deductive approach during theme formulation as we aimed to identify themes that could easily be transferred into prescriptive knowledge. Particularly, we map limitations voiced by the participants onto *material* aspects of the design artifact, meaning structural aspects that support or hinder the artifacts design goals. Interviewees wishes and suggestions as well as potential use cases give us insights about the AI energy label's *action* properties by prescribing what the label does or should do to address user needs. Finally, themes describing the use context of AI energy labels give us intelligence about the possible *boundary conditions* under which the AI energy label design functions best. In the final round, we use selective coding in an effort to unify the themes. Making use of our conceptualization of artifact action, materiality, and boundary conditions, we formulate a final code set. The four dimensions harmonize limitations and improvement suggestions with the different use contexts. Although the theorizing process is presented in a sequential way for the sake of simplicity, the analysis was iteratively performed,

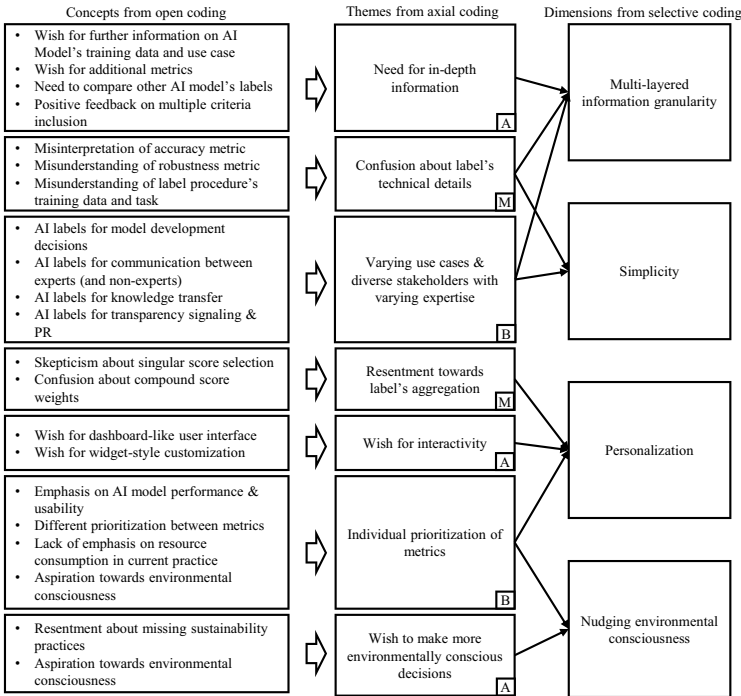


Fig. 3. Code structure resulting from grounded theorizing of the interviews marked with themes relating to materiality (M), action (A), and boundary conditions (B) [4].

complying with grounded theory methodology. Our final set of concepts, themes, and dimensions is displayed in Fig. 3. The code system is available as a dataset.¹

4 Design Principles for AI Energy Labels

Our analysis resulted in four DPs providing prescriptive knowledge on how future AI energy labels should be refined and designed. We follow the format for consistent and precise DPs by Chandra et al. [4] as a textual skeleton, which we introduced in Sect. 2.1.

Our first DP addresses the balancing act between catering to some interviewees need for in-depth information and at the same time lowering the confusion that others voiced with regards to technical details and the labeling procedure. On the one hand, some interviewees suggested label elements with additional information, like I7, who proposed to optionally show “a larger secondary label for those who are interested in showing other relevant information, perhaps comparable models” (p.150). On the other hand, participants voiced misapprehensions about labels’ technical details and see “the disadvantage that you need a lot of background knowledge” (I6, p.190). To meet the various levels of AI expertise of users that could potentially interact with the AI energy label, the information granularity should be switchable between different layers. For example, QR codes can be used to enable the search for more technical information. We therefore propose the first DP:

DP 1. *Accompany labels with a multi-layered information framework featuring navigable and ordered layers that offer information with varied complexity and depth. This structure enables users to access detailed technical insights if desired while ensuring that less experienced users can engage with simplified representations aligned with their level of expertise.*

In the interview study, there was evidence that some practitioners (e.g., non-technical stakeholders, or non-expert-developers using AI-as-a-Service) are not able to interpret basic technical details, like the accuracy metric. As such, the most abstract layer of the multi-layered framework should provide users with the option to understand metrics without measurement units, allowing them to learn about AI model behavior at an abstract glance, but also offering in-depth details on demand. Certain use cases of the AI energy labels like transfer of knowledge and advertisement also require a simplistic label representation, which leads us into the second DP:

DP 2. *On the most abstract layer of the multi-layered framework described in DP1, present only high-level labels devoid of technical metrics (e.g., accuracy or robustness) to prevent overwhelming or misleading users who have limited AI expertise.*

By using simplified indicators and visual aids such as icons and colors that convey information in an intuitive way, AI model characteristics can be communicated

¹ <https://github.com/alex-vdst/DESRIST25-reflective-theorizing/>.

at a more comprehensible level. I11 praised the simplistic aspect of the AI energy label, stating “you can always pick people up with categories like red, yellow, and green, in my experience” (p. 106). However, it is important to determine a color scale that meets accessibility guidelines, for example with regards to color vision deficiency.

Relating to the aggregation of metrics on the AI energy label, participants voiced their skepticism about the compound score that is displayed most prominently on the label. Confusion was expressed with regards to the different singular metrics that went into the compound score as well as the weights used for aggregation. Additionally, participants raised concerns about the highly individual prioritization of metrics between different companies and users, stating “the A that (the compound score) shows here is misleading, because if accuracy is totally important to me, then this is definitely the wrong model” (I16, p. 114). As a solution for these issues, participants expressed their wish for interactive labels that allowed for customization of scores and score weights based on individual preferences. To account for this feedback, we formulate the third DP:

DP 3. *Use fair defaults for balancing all performance aspects but also allow for a personalized experience and compound model scoring via an interactive interface, to enable users to cater to individual needs and use case requirements.*

Lastly, DP4 stands out as particularly meaningful, being synthesized for its unique ability to address a critical purpose of the label: Enabling environmentally-conscious model selection and thus fostering sustainable development of (and with) AI. For some interviewees, inner conflicts regarding the trade-off between AI model performance and energy consumption were observed. They firstly stated that predictive performance metrics like accuracy are the most important factors in AI decision making in their current practice. However, upon analyzing the AI energy labels and seeing the energy consumption score, they indicated their wish to make decisions between AI models that are more environment-friendly. This leads us to our fourth and final DP:

DP 4. *Provide any model information with prominently displayed resource consumption scores using clear visual cues, such as green icons, in order to nudge developers toward sustainable AI model use.*

The open-ended style of the interview questions allowed for rich reflections from users regarding the IT artifact, providing valuable insights into their experiences and needs. By combining grounded theory with the structured components of DPs, we were able to translate these reflections into DPs that we deem meaningful and actionable.

5 Discussion, Limitations, and Future Work

5.1 Discussion

In alignment with the two RQs that we posed, the findings of our study are twofold. First, we provide context on the current state of the emerging DSR

project type of *reflective studies* that contribute to design theory without simultaneous construction of artifacts [31]. We address the limitations of *ex-situ studies*, particularly their lack of user involvement during theorizing, by introducing *user interviews* as a key method for gaining insights into IT artifacts. These interviews allow us to capture users' perspectives on principles of action and materiality, providing an *in-situ* understanding of the artifact's impact. To transform these insights into prescriptive knowledge, we leverage *grounded theory methodology*, which offers rigorous and actionable procedures for synthesizing qualitative data, especially when combining it with an established DP formalization framework [4]. With our work, we extend the existing range of methods for DSR studies, opening up more opportunities for design researchers to reflectively generate meaningful and use-context-aware design theory from existing artifacts. In this vein, we challenge the apparent incompatibility between grounded theory and design science projects that prevails within the DSR community [14]. As we implied in Sect. 2.3, we resolve the seeming clash between problem-solution oriented DSR and world-describing grounded theory by applying the latter within the context of emerging reflective theorizing studies that do not focus on constructing problem-solving artifacts.

Implementing this into practice, we apply our qualitative reflective theorizing approach on AI energy labels, making up the second part of this research's contribution. Via grounded theorizing on qualitative data gained from AI user and developer interviews, we formulate four DPs that inform the efficient design of AI energy labels. We coded and categorized users' sentiments and opinions toward different components of the label before unifying them with the four final themes, which our DPs describe. We find that labels should be obtained from a *multi-layered framework*, to provide information tailored to user expertise, with the most abstract layer showing only *simplistic, non-technical information* to avoid confusion. An *interactive interface* should allow users to personalize the compound scoring, aligning the impact of different performance metrics with their individual priorities. Additionally, *resource consumption scores* should be prominently displayed to promote environmentally sustainable decisions. Using qualitative reflective theory, we were able to transfer expressed limitations and opportunities of AI energy labels into actionable design knowledge that could inform the development of sophisticated AI energy labels by practitioners and policy-makers. By addressing AI energy label users' needs for multi-granular information layers, personalization, and simplistic representations, future AI energy label designers can facilitate effective information sharing and informed decision making regarding AI models. These principles could also pave the way for AI energy labels to become an industry standard, promoting consistency and widespread adoption in sustainable AI practices. Finally, we translated AI users' shifts in sustainability prioritization, observed upon exposure to the label, into an actionable DP. This descriptive knowledge contributes to a deeper understanding of how AI energy labels can drive sustainable behavior among AI users, providing a localized solution for environmentally conscious AI model choices. In

doing so, we address the broader global challenge of mitigating the environmental impact of AI development.

5.2 Limitations and Future Work

It is important to acknowledge the limitations of our study. The final DPs we derived were founded in a rich body of interview data and generated in a rigorous grounded theorizing process. We therefore view the DPs to be sufficiently validated via inductive reasoning in the similar way that grounded theory in other bodies of literature produces validated theory, models or rich descriptions [33]. However, we recognize that the evaluation of our DPs may not fully adhere to the standard expected in other types of DSR studies. Future research should investigate evaluation strategies for reflectively derived DPs, emphasizing DP instantiation and the question of *who* should be responsible for the evaluation, by differentiating between the different roles that humans can take within theorizing [12, 22]. For instance, a meaningful evaluation strategy could involve validating the generated DPs with the original interviewees.

The open-ended interviews that we analyzed yielded valuable but varied insights. Future research should explore structured interview techniques like thinking-aloud protocols, narrative interviews, or stream-of-consciousness methods to capture richer reflections of users with regards to IT artifacts. Developing a framework with interview guidelines could enhance consistency while supporting flexible, user-based reflective theorizing. Additionally, other qualitative and quantitative approaches such as focus groups or eye-tracking should be evaluated to capture richer reflections.

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